Tangent Lévy v2
29 August 2025 Historical-Diffusion and Q-Drift Audit

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Abstract

This note records a deliberately misspecified Tangent Lévy experiment initialized on the same 29 August 2025 SPX session used by the SDLV study in `main2.tex`. The double-exponential static family fits the market poorly and no retained historical reconstruction passes the production gate. Nevertheless, a target-local 16-factor diffusion is estimated from historical codebook changes and evolved under the convolutional Tangent Lévy HJM drift. The requested unconditional and strategy-conditional Q diagnostics are then applied to 100,000 simulated paths in the paper's restricted universe of spot plus 44 fixed-cash-strike options. The no-arbitrage conclusion is reported separately from, and does not rehabilitate, the failed market fit.

1 Data and initialization

The initial state is calibrated to the SPX option book dated 29 August 2025. The book contains 4,036 retained call-equivalent bid/ask observations over 32 expiries. Historical codebooks span 3 January 2022 through 31 December 2024. Of 729 requested sessions, 726 are retained and 698 adjacent transitions are valid one-session covariance observations. Three sessions are excluded by the unchanged quote-book integrity gate: 10 May 2023, 17 November 2023, and 11 December 2023. No crossed or otherwise invalid observation is repaired.

The target spot and all option values are generated consistently from the Tangent Lévy codebook. Spot follows the compensated compound-Poisson law implied by the current short-end Lévy density. Options are repriced from the simulated forward density with a lattice-consistent compound-Poisson FFT.

2 Static calibration result

The independent per-expiry optimization converges, but the family has poor capacity for this surface. Only 1,375 of 4,036 observations (34.07%) lie inside bid/ask. The excess-price RMSE is 5.202 quoted half-spreads. Enforcing calendar consistency leaves 898 observations (22.25%) inside bid/ask and an RMSE of 6.704 half-spreads. Reconstructing the 71-node daily forward-density grid leaves 224 observations (5.55%) inside bid/ask and an RMSE of 25.427 half-spreads.

The target reconstruction requires relative L^1 modification 1.0498 and has a repricing RMSE of 6.938 half-spreads, so it is a **FAIL** against the pre-registered reconstruction gate. All 726 historical states fail the same gate. This is the expected model-capacity result, not an optimizer failure.

3 Historical diffusion and measures

Daily forward-density changes are centered and projected into the discrete mass and exponential-moment constraint nullspace. Admissible coordinates are selected relative to the 29 August 2025

initial density, while covariance is estimated solely from the 698 historical one-session transitions. Sixteen factors retain 95.002% of projected variance over 142 active coordinates out of 21,371. The maximum factor-constraint residual is 3.23×10^{-17}.

Under Q, the volatility fields determine the convolutional HJM drift. A separate constant market-price-of-risk regression estimates the descriptive P drift in the same factor span; it explains 99.995% of the fitted target drift with residual RMSE 210.206 in codebook-drift units. This P regression is not used in either no-arbitrage test.

The Gaussian factor system otherwise reaches the positivity boundary almost immediately. Simulation therefore applies a predictable global scalar time change: factor loadings are multiplied by γ, their quadratic HJM drift by γ^2, and the P risk-premium term (when used) by γ. The scale is fixed before each shock and reserves twelve conditional standard deviations of coordinate-wise headroom. Density clipping, shock resampling, and path deletion are forbidden. The final γ diagnostics are essential to interpreting the drift tests.

4 Risk-neutral audit design

The audit follows `main2.tex`: 100,000 paths form 50,000 independent antithetic pairs; each path covers 20 business days with eight substeps per day. The trading universe is spot and eleven initially out-of-the-money fixed-cash-strike options at each of 5, 10, 15, and 20 business-day expiries. For each live contract the tested one-day return is the value change divided by pre-step spot.

The unconditional diagnostic first averages every antithetic pair and then forms a studentized mean for each live day–instrument coordinate. The two conditional diagnostics fit uncapped affine-plus-width-32 SiLU policies on one pair-preserving fold and evaluate their predictable portfolio return on the opposite fold. One uses the squared objective and the other the entropy objective specified in `main2.tex`. A conditional test passes at the 5% two-sided level when its out-of-fold 95% interval contains zero.

5 Results

All 100,000 paths complete without clipping, deletion, or a frozen path. The time change is nevertheless dominant: its minimum is 2.13×10^{-10} and its mean is 0.05452. Of 16,000,000 path–substeps, 15,858,939 (99.12%) are boundary-limited, 14,182,352 have $\gamma < 0.1$, 12,516,601 have $\gamma < 0.01$, and 9,211,196 have $\gamma < 10^{-4}$.

The unconditional screen is a clear **FAIL**. Among all 570 live coordinates, the maximum absolute z statistic is 40.632 and the 99th percentile is 28.089. There are 276 coordinates above 1.96, 229 above 2.576, and 134 above 5. These counts are far beyond the ordinary multiple-testing frequency under a standard-normal coordinate null.

Conditional test	Mean score	Pair SE	t statistic	95% interval	Result
Squared	1.770×10^{-7}	4.022×10^{-8}	4.400	$[0.981, 2.558] \times 10^{-7}$	**FAIL**
Entropy	3.478×10^{-8}	4.914×10^{-9}	7.078	$[2.515, 4.442] \times 10^{-8}$	**FAIL**

Table 1: Pair-preserving, two-fold out-of-fold conditional drift tests.

Both strategy tests reject zero predictable drift at conventional levels: each complete held-out confidence interval is positive. The squared test has $t = 4.400$ and the entropy test has $t = 7.078$.

6 Interpretation

The stochastic 29 August 2025 construction does *not* pass the requested Q no-arbitrage diagnostics. This is not evidence that Tangent Lévy theory is arbitrageable. It is evidence that the present empirical/discretized construction—a very poor static reconstruction, historical factors defined on only 142 target-admissible coordinates, and an almost always active global boundary time change—does not preserve the fixed-contract martingale identities closely enough for the 100,000-path test.

The predictable time change preserves the continuous-time HJM scaling algebraically, but it does not cure finite-step transport/pricing bias, and its near-freezing makes the fitted historical diffusion economically remote from the unscaled estimate. Accordingly, neither the failed drift test nor any future non-rejection should be generalized beyond this restricted universe and strategy class. The correct empirical conclusion is that this stochastic specification is unsuitable as a risk-neutral model in its current form.

7 Retained artifacts

The canonical external result root is `/home/hans/quantdata_data/training/dysanos/tangent-levy/v2/spx-2025-08-29-historical-diffusion-v2`. It contains the target and historical checkpoints, calibration and measure reports, the canonical Q population, unconditional audit, conditional JSON reports, and resumable conditional-policy checkpoints. Large parallel-generation chunks are deleted after the canonical arrays and audit are verified.